

Flip-flop is a 1 bit memory cell which can be used for storing the digital data. To increase the storage capacity in terms of number of bits, we have to use a group of flip-flop. Such a group of flip-flop is known as a **Register**. The **n-bit register** will consist of **n** number of flip-flop and it is capable of storing an **n-bit** word.

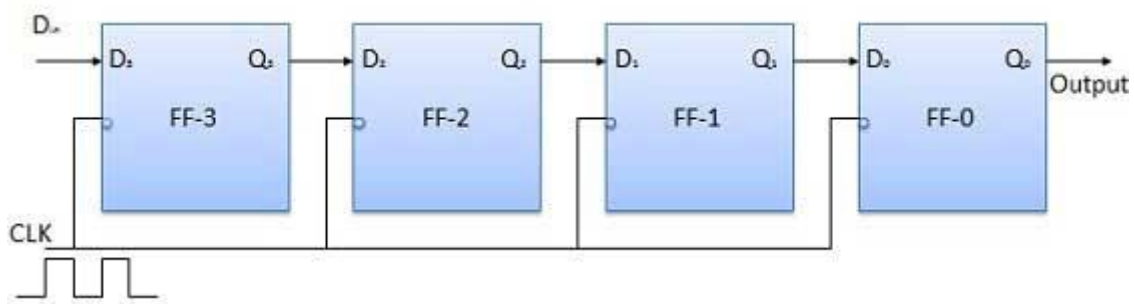
The binary data in a register can be moved within the register from one flip-flop to another. The registers that allow such data transfers are called as **shift registers**. There are four mode of operations of a shift register.

- Serial Input Serial Output
- Serial Input Parallel Output
- Parallel Input Serial Output
- Parallel Input Parallel Output

Serial Input Serial Output

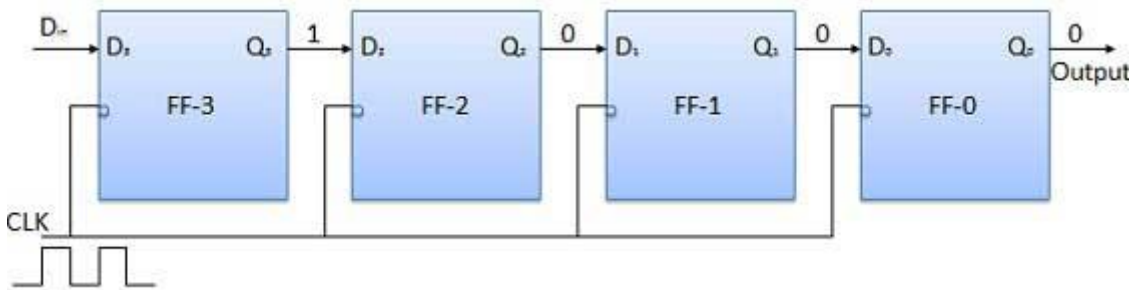
Let all the flip-flop be initially in the reset condition i.e. $Q_3 = Q_2 = Q_1 = Q_0 = 0$. If an entry of a four bit binary number 1 1 1 1 is made into the register, this number should be applied to D_{in} bit with the LSB bit applied first. The D input of FF-3 i.e. D_3 is connected to serial data input D_{in} . Output of FF-3 i.e. Q_3 is connected to the input of the next flip-flop i.e. D_2 and so on.

Block Diagram

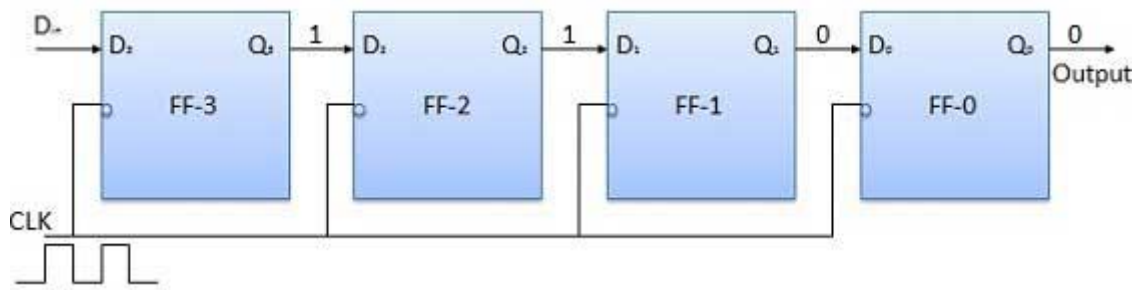


Operation

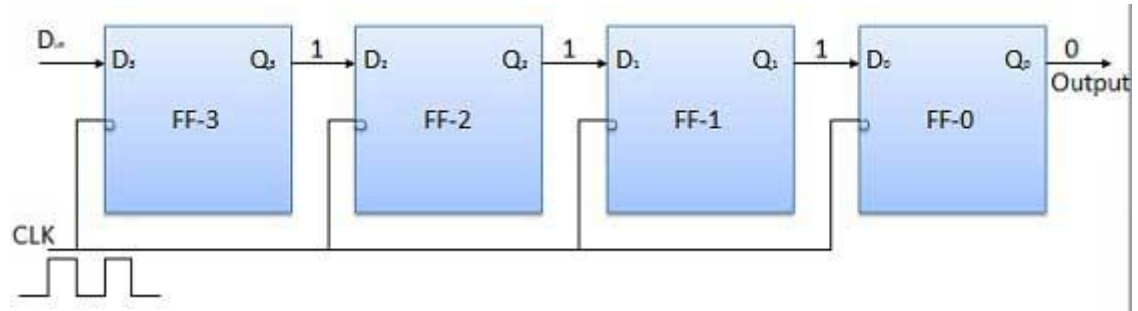
Before application of clock signal, let $Q_3 Q_2 Q_1 Q_0 = 0000$ and apply LSB bit of the number to be entered to D_{in} . So $D_{in} = D_3 = 1$. Apply the clock. On the first falling edge of clock, the FF-3 is set, and stored word in the register is $Q_3 Q_2 Q_1 Q_0 = 1000$.



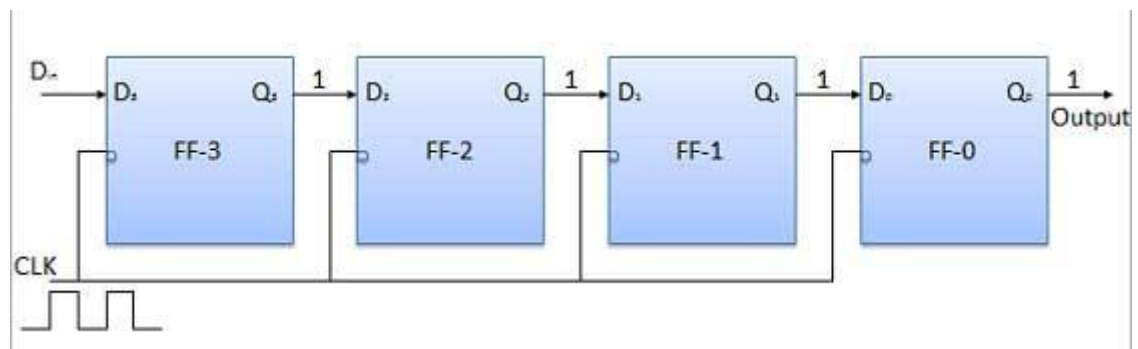
Apply the next bit to D_{in} . So $D_{in} = 1$. As soon as the next negative edge of the clock hits, FF-2 will set and the stored word change to $Q_3 Q_2 Q_1 Q_0 = 1100$.



Apply the next bit to be stored i.e. 1 to D_{in} . Apply the clock pulse. As soon as the third negative clock edge hits, FF-1 will be set and output will be modified to $Q_3 Q_2 Q_1 Q_0 = 1110$.



Similarly with $D_{in} = 1$ and with the fourth negative clock edge arriving, the stored word in the register is $Q_3 Q_2 Q_1 Q_0 = 1111$.

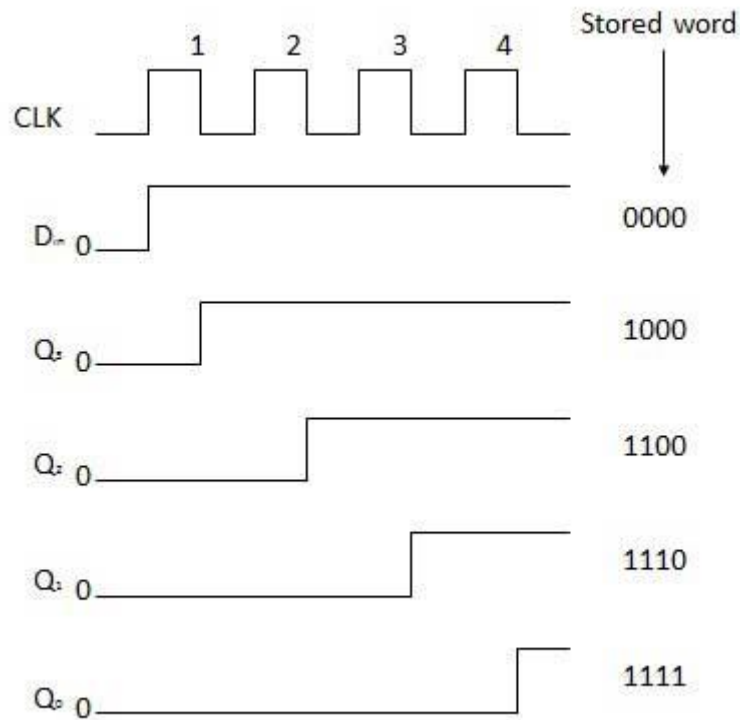


Truth Table

	CLK	$D_{in} = Q_3$	$Q_3 = D_2$	$Q_2 = D_1$	$Q_1 = D_0$	Q_0
Initially			0	0	0	0
(i)	↓	1	1	0	0	0
(ii)	↓	1	1	1	0	0
(iii)	↓	1	1	1	1	0
(iv)	↓	1	1	1	1	1

→ Direction of data travel

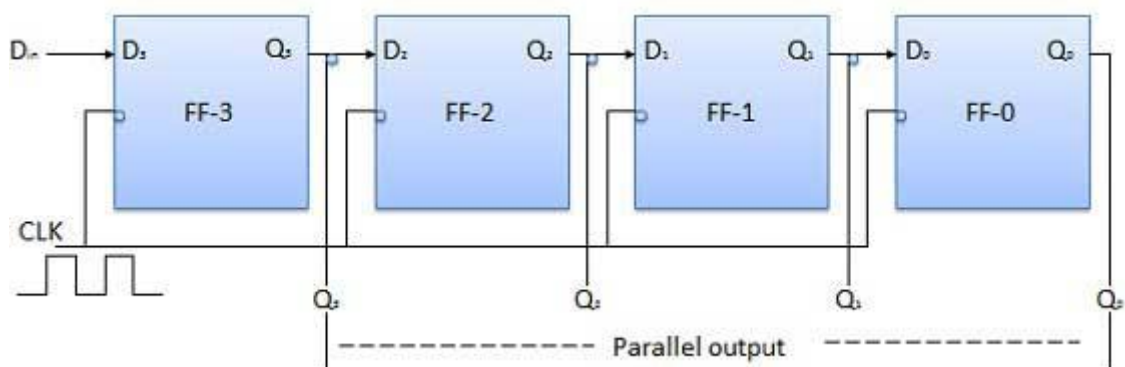
Waveforms



Serial Input Parallel Output

- In such types of operations, the data is entered serially and taken out in parallel fashion.
- Data is loaded bit by bit. The outputs are disabled as long as the data is loading.
- As soon as the data loading gets completed, all the flip-flops contain their required data, the outputs are enabled so that all the loaded data is made available over all the output lines at the same time.
- 4 clock cycles are required to load a four bit word. Hence the speed of operation of SIPO mode is same as that of SISO mode.

Block Diagram



Parallel Input Serial Output (PISO)

- Data bits are entered in parallel fashion.
- The circuit shown below is a four bit parallel input serial output register.

- Output of previous Flip Flop is connected to the input of the next one via a combinational circuit.
- The binary input word B_0, B_1, B_2, B_3 is applied through the same combinational circuit.
- There are two modes in which this circuit can work namely - shift mode or load mode.

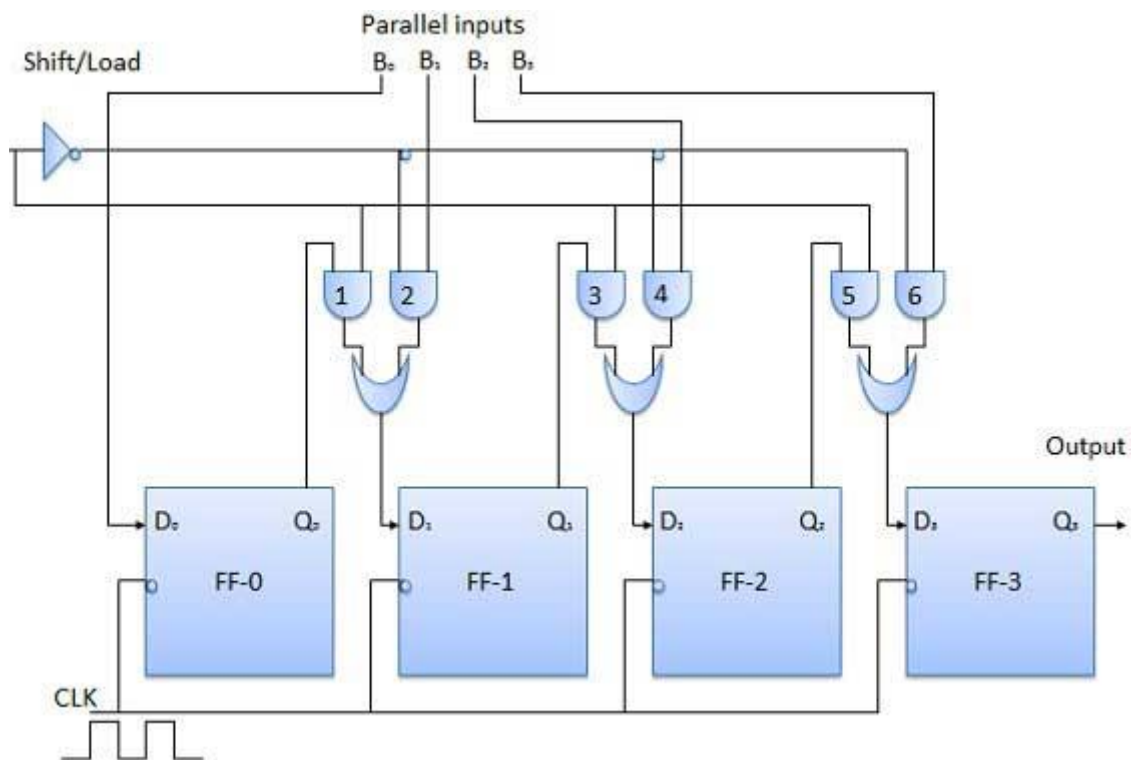
Load mode

When the shift/load bar line is low (0), the AND gate 2, 4 and 6 become active they will pass B_1, B_2, B_3 bits to the corresponding flip-flops. On the low going edge of clock, the binary input B_0, B_1, B_2, B_3 will get loaded into the corresponding flip-flops. Thus parallel loading takes place.

Shift mode

When the shift/load bar line is high (1), the AND gate 2, 4 and 6 become inactive. Hence the parallel loading of the data becomes impossible. But the AND gate 1, 3 and 5 become active. Therefore the shifting of data from left to right bit by bit on application of clock pulses. Thus the parallel in serial out operation takes place.

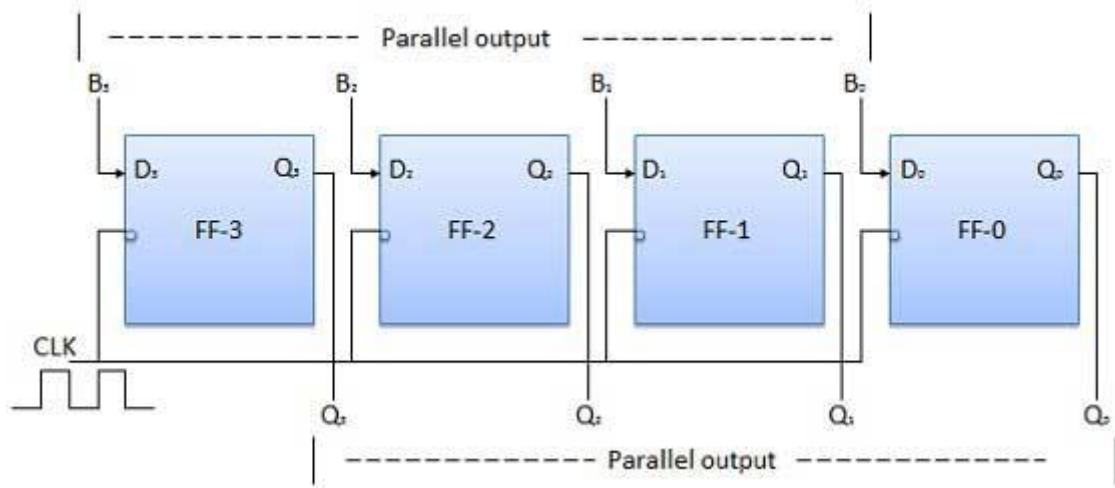
Block Diagram



Parallel Input Parallel Output (PIPO)

In this mode, the 4 bit binary input B_0, B_1, B_2, B_3 is applied to the data inputs D_0, D_1, D_2, D_3 respectively of the four flip-flops. As soon as a negative clock edge is applied, the input binary bits will be loaded into the flip-flops simultaneously. The loaded bits will appear simultaneously to the output side. Only clock pulse is essential to load all the bits.

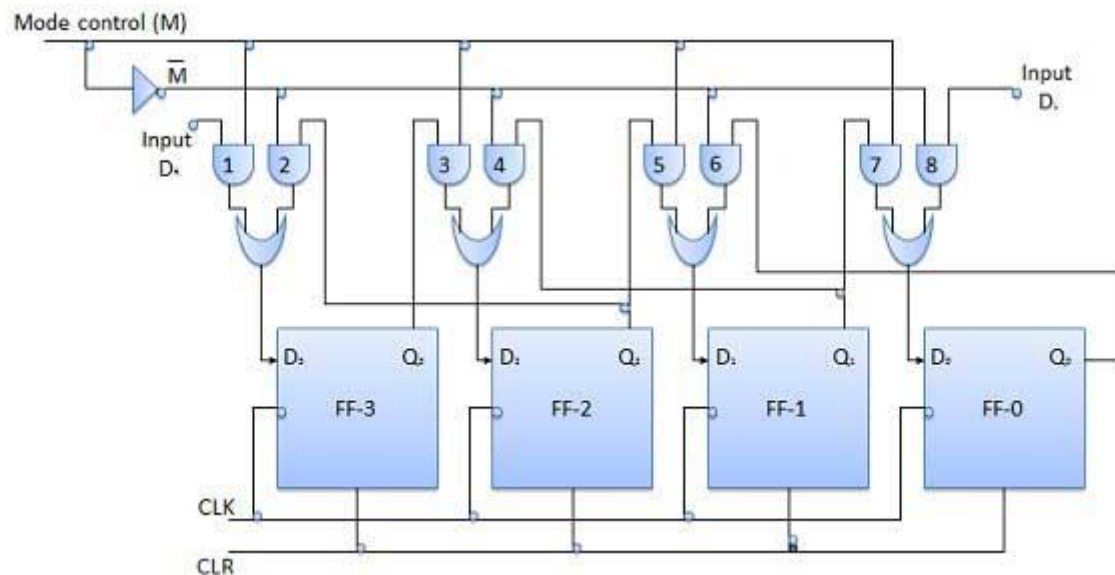
Block Diagram



Bidirectional Shift Register

- If a binary number is shifted left by one position then it is equivalent to multiplying the original number by 2. Similarly if a binary number is shifted right by one position then it is equivalent to dividing the original number by 2.
- Hence if we want to use the shift register to multiply and divide the given binary number, then we should be able to move the data in either left or right direction.
- Such a register is called bi-directional register. A four bit bi-directional shift register is shown in fig.
- There are two serial inputs namely the serial right shift data input DR, and the serial left shift data input DL along with a mode select input (M).

Block Diagram



Operation

S.N.	Condition	Operation

1	With M = 1 – Shift right operation	If $M = 1$, then the AND gates 1, 3, 5 and 7 are enabled whereas the remaining AND gates 2, 4, 6 and 8 will be disabled. The data at D_R is shifted to right bit by bit from FF-3 to FF-0 on the application of clock pulses. Thus with $M = 1$ we get the serial right shift operation.
2	With M = 0 – Shift left operation	When the mode control M is connected to 0 then the AND gates 2, 4, 6 and 8 are enabled while 1, 3, 5 and 7 are disabled. The data at D_L is shifted left bit by bit from FF-0 to FF-3 on the application of clock pulses. Thus with $M = 0$ we get the serial left shift operation.

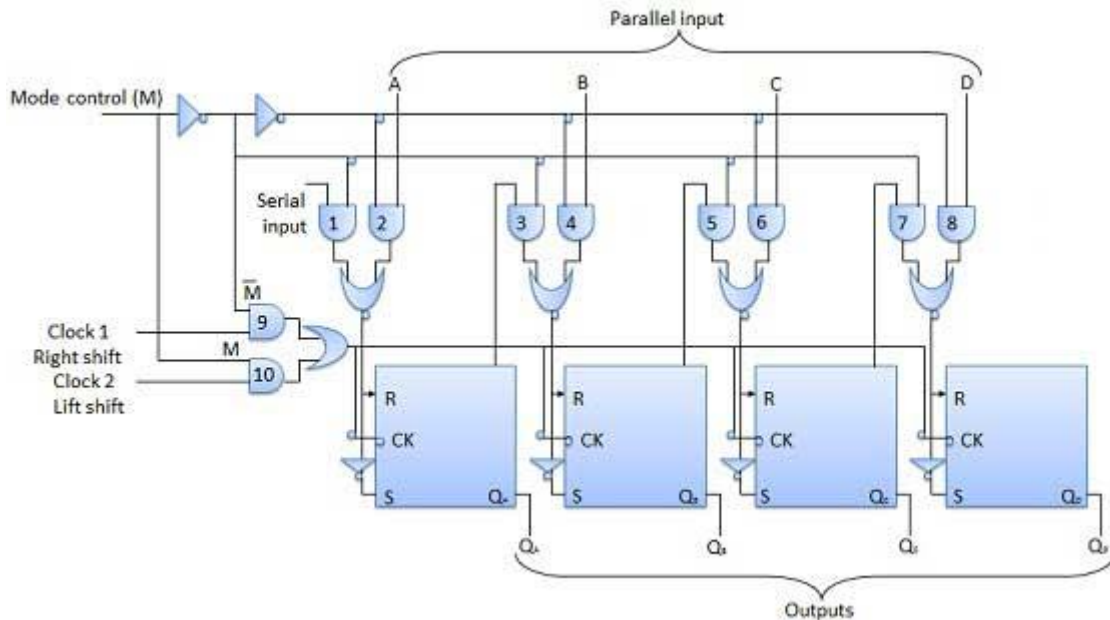
Universal Shift Register

A shift register which can shift the data in only one direction is called a uni-directional shift register. A shift register which can shift the data in both directions is called a bi-directional shift register. Applying the same logic, a shift register which can shift the data in both directions as well as load it parallelly, is known as a universal shift register. The shift register is capable of performing the following operation –

- Parallel loading
- Left Shifting
- Right shifting

The mode control input is connected to logic 1 for parallel loading operation whereas it is connected to 0 for serial shifting. With mode control pin connected to ground, the universal shift register acts as a bi-directional register. For serial left operation, the input is applied to the serial input which goes to AND gate-1 shown in figure. Whereas for the shift right operation, the serial input is applied to D input.

Block Diagram



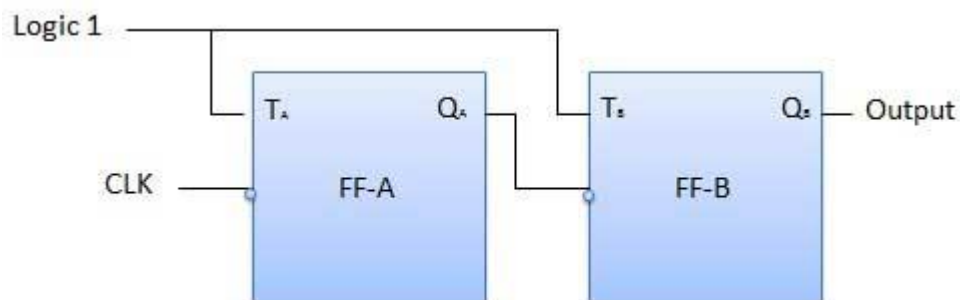
Counter is a sequential circuit. A digital circuit which is used for a counting pulses is known counter. Counter is the widest application of flip-flops. It is a group of flip-flops with a clock signal applied. Counters are of two types.

- Asynchronous or ripple counters.
- Synchronous counters.

Asynchronous or ripple counters

The logic diagram of a 2-bit ripple up counter is shown in figure. The toggle (T) flip-flop are being used. But we can use the JK flip-flop also with J and K connected permanently to logic 1. External clock is applied to the clock input of flip-flop A and Q_A output is applied to the clock input of the next flip-flop i.e. FF-B.

Logical Diagram



Operation

S.N.	Condition	Operation
1	Initially let both the FFs be in the	$Q_B Q_A = 00$ initially

	reset state	
2	After 1st negative clock edge	As soon as the first negative clock edge is applied, FF-A will toggle and Q_A will be equal to 1. Q_A is connected to clock input of FF-B. Since Q_A has changed from 0 to 1, it is treated as the positive clock edge by FF-B. There is no change in Q_B because FF-B is a negative edge triggered FF. $Q_B Q_A = 01$ after the first clock pulse.
3	After 2nd negative clock edge	On the arrival of second negative clock edge, FF-A toggles again and $Q_A = 0$. The change in Q_A acts as a negative clock edge for FF-B. So it will also toggle, and Q_B will be 1. $Q_B Q_A = 10$ after the second clock pulse.
4	After 3rd negative clock edge	On the arrival of 3rd negative clock edge, FF-A toggles again and Q_A become 1 from 0. Since this is a positive going change, FF-B does not respond to it and remains inactive. So Q_B does not change and continues to be equal to 1. $Q_B Q_A = 11$ after the third clock pulse.
5	After 4th negative clock edge	On the arrival of 4th negative clock edge, FF-A toggles again and Q_A becomes 0 from 1. This negative change in Q_A acts as clock pulse for FF-B. Hence it toggles to change Q_B from 1 to 0. $Q_B Q_A = 00$ after the fourth clock pulse.

Truth Table

Clock	Counter output		State number	Decimal Counter output
	Q_B	Q_A		
Initially	0	0	—	0
1st	0	1	1	1
2nd	1	0	2	2
3rd	1	1	3	3
4th	0	0	4	0

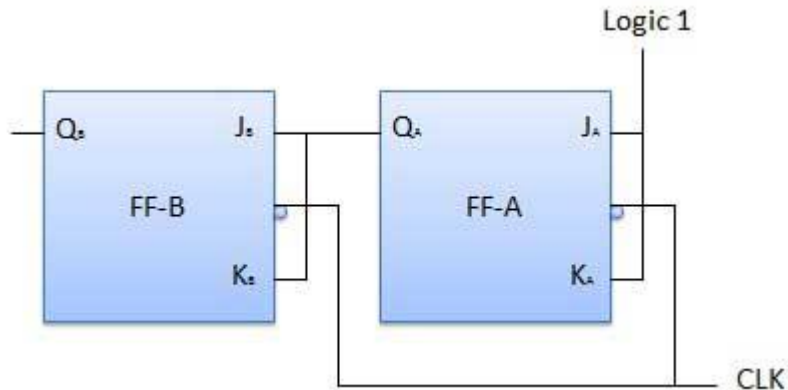
Synchronous counters

If the "clock" pulses are applied to all the flip-flops in a counter simultaneously, then such a counter is called as synchronous counter.

2-bit Synchronous up counter

The J_A and K_A inputs of FF-A are tied to logic 1. So FF-A will work as a toggle flip-flop. The J_B and K_B inputs are connected to Q_A .

Logical Diagram



Operation

S.N.	Condition	Operation
1	Initially let both the FFs be in the reset state	$Q_B Q_A = 00$ initially.
2	After 1st negative clock edge	As soon as the first negative clock edge is applied, FF-A will toggle and Q_A will change from 0 to 1. But at the instant of application of negative clock edge, Q_A, $J_B = K_B = 0$. Hence FF-B will not change its state. So Q_B will remain 0. $Q_B Q_A = 01$ after the first clock pulse.
3	After 2nd negative clock edge	On the arrival of second negative clock edge, FF-A toggles again and Q_A changes from 1 to 0. But at this instant Q_A was 1. So $J_B = K_B = 1$ and FF-B will toggle. Hence Q_B changes from 0 to 1. $Q_B Q_A = 10$ after the second clock pulse.

4	After 3rd negative clock edge	On application of the third falling clock edge, FF-A will toggle from 0 to 1 but there is no change of state for FF-B. $Q_B Q_A = 11$ after the third clock pulse.
5	After 4th negative clock edge	On application of the next clock pulse, Q_A will change from 1 to 0 as Q_B will also change from 1 to 0. $Q_B Q_A = 00$ after the fourth clock pulse.

Classification of counters

Depending on the way in which the counting progresses, the synchronous or asynchronous counters are classified as follows –

- Up counters
- Down counters
- Up/Down counters

UP/DOWN Counter

Up counter and down counter is combined together to obtain an UP/DOWN counter. A mode control (M) input is also provided to select either up or down mode. A combinational circuit is required to be designed and used between each pair of flip-flop in order to achieve the up/down operation.

- Type of up/down counters
- UP/DOWN ripple counters
- UP/DOWN synchronous counter

UP/DOWN Ripple Counters

In the UP/DOWN ripple counter all the FFs operate in the toggle mode. So either T flip-flops or JK flip-flops are to be used. The LSB flip-flop receives clock directly. But the clock to every other FF is obtained from ($Q = Q \text{ bar}$) output of the previous FF.

- **UP counting mode (M=0)** – The Q output of the preceding FF is connected to the clock of the next stage if up counting is to be achieved. For this mode, the mode select input M is at logic 0 (M=0).
- **DOWN counting mode (M=1)** – If M = 1, then the Q bar output of the preceding FF is connected to the next FF. This will operate the counter in the counting mode.

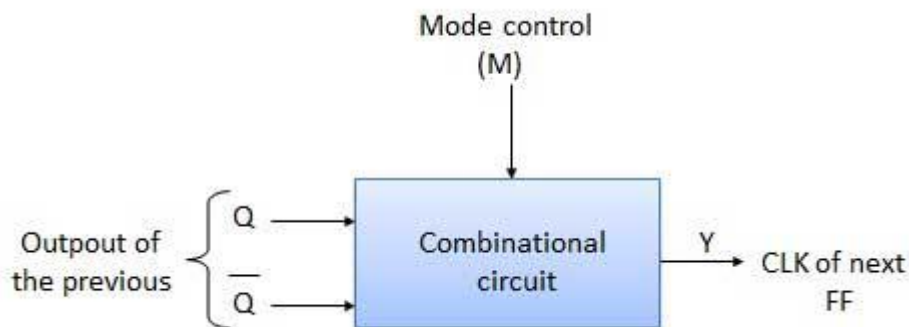
Example

3-bit binary up/down ripple counter.

- 3-bit – hence three FFs are required.
- UP/DOWN – So a mode control input is essential.

- For a ripple up counter, the Q output of preceding FF is connected to the clock input of the next one.
- For a ripple up counter, the Q output of preceding FF is connected to the clock input of the next one.
- For a ripple down counter, the Q bar output of preceding FF is connected to the clock input of the next one.
- Let the selection of Q and Q bar output of the preceding FF be controlled by the mode control input M such that, If M = 0, UP counting. So connect Q to CLK. If M = 1, DOWN counting. So connect Q bar to CLK.

Block Diagram



Truth Table

Inputs			Outputs
M	Q	$\overline{Q}$	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

For M = 0, Y = Q (Up counter)
 For M = 1, Y = $\overline{Q}$ (Down counter)

Operation

S.N.	Condition	Operation
1	Case 1 – With M = 0 (Up counting mode)	If M = 0 and M bar = 1, then the AND gates 1 and 3 in fig. will be enabled whereas the AND gates 2 and 4 will be disabled. Hence Q_A gets connected to the clock input of FF-B and Q_B gets connected to the clock input of FF-C.

		These connections are same as those for the normal up counter. Thus with $M = 0$ the circuit work as an up counter.
2	Case 2: With $M = 1$ (Down counting mode)	If $M = 1$, then AND gates 2 and 4 in fig. are enabled whereas the AND gates 1 and 3 are disabled. Hence Q_A bar gets connected to the clock input of FF-B and Q_B bar gets connected to the clock input of FF-C. These connections will produce a down counter. Thus with $M = 1$ the circuit works as a down counter.

Modulus Counter (MOD-N Counter)

The 2-bit ripple counter is called as MOD-4 counter and 3-bit ripple counter is called as MOD-8 counter. So in general, an n-bit ripple counter is called as modulo-N counter. Where, MOD number = 2^n .

Type of modulus

- 2-bit up or down (MOD-4)
- 3-bit up or down (MOD-8)
- 4-bit up or down (MOD-16)

Application of counters

- Frequency counters
- Digital clock
- Time measurement
- A to D converter
- Frequency divider circuits
- Digital triangular wave generator.